

Post-irradiation annealing of neutron-irradiated EUROFER97

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ABSTRACT

Recent post-irradiation annealing (PIA) investigations on the neutron-irradiated reduced-activation ferritic-martensitic (RAFM) steel EUROFER97 have led to an almost complete recovery of the mechanical properties. To identify responsible microstructural processes, transmission electron microscopy (TEM) investigations on neutron-irradiated (330 °C, 15 dpa) and subsequently annealed (550 °C, 3 h) EUROFER97 samples have been carried out. It is observed that PIA results in a reduction of dislocation loops density with a simultaneous increase in their mean size and formation of dislocation lines/networks. Additionally, a reduction of cavities density with increment in their mean size was observed. The loops' annealing behaviour appears to be vacancy/interstitial diffusion driven and is situated between the two extremes of isolated loops shrinkage/annihilation and their coarsening. Out of the two phenomena, the one that transpires depend on the presence or absence of the external sources and sinks in the loops neighbourhood. These competing mechanisms define the overall loops size distribution evolution upon PIA. Additionally, vacancy or their cluster migration and coalescence could be the main mechanism for cavities coarsening. The observed microstructural recovery upon PIA explains the recovery of mechanical properties by healing of irradiation induced defects.

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1. Introduction

Reduced-activation ferritic-martensitic (RAFM) steel EUROFER97 is a candidate structural material for the First Wall of nuclear fusion reactor DEMO [1]. This advanced nuclear reactor will impose harsh working conditions such as high doses of neutron irradiation which is known to degrade mechanical properties strongly [2,3]. For an example, irradiation doses of up to 32 dpa at 300–340 °C have been shown to induce large low-temperature hardening and embrittlement. This is mainly attributed to the neutron irradiation induced microstructural defects, such as dislocation loops, cavities, solute clustering, precipitates and grain boundary segregation [4–8]. In general, dislocation loops and cavities are considered to be the main source of hardening [4,5]. In EUROFER97, while cavities are the aggregates of vacancies (sometimes stabilized with He), loops are prismatic in nature that form as a result of clustering of the disc-shaped platelets of interstitials. Loops' Burgers vector b is perpendicular to the line direction of the loop periphery [9] and two type of loops with Burgers vector $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ have been confirmed to exist in neutron-irradiated EUROFER97 [5,7].

Temperature plays an important role in the evolution of irradiation induced defects [10,11]. For instance, Yao et al. [10] performed

in-situ ion-irradiation experiments on pure α -iron and showed that the small mobile $\frac{1}{2}\langle 111 \rangle$ loops were subsumed by sessile $\langle 100 \rangle$ loops at temperatures above 450 °C, while at 500 °C, $\langle 100 \rangle$ loops developed into large networks, and no visible $\frac{1}{2}\langle 111 \rangle$ loops formed [10]. Furthermore, Arakawa et al. [12–15] also showed various effects of temperature on irradiation induced defects in pure α -iron. Therefore, a proper post-irradiation temperature exposure can be a viable approach to favourably modify neutron damage; and thus, recovering mechanical properties. Indeed, several studies in the past have shown the importance of post-irradiation annealing (PIA) in recovering hardening and embrittlement behaviour of various irradiated austenitic stainless steels [16–24]. Recently, PIA of neutron-irradiated EUROFER97 (15 and 70 dpa at 330 °C) at 550 °C for 3 h has also yielded in a substantial reduction of the ductile to brittle transition temperature (DBTT) as well as a nearly complete recovery of the low-temperature hardening [3,25,26]. Understanding microstructural mechanisms behind such recovery is of high scientific interest. In addition, provided that underlying processes are well understood, the application of recovery heat treatment to the in-vessel components of future fusion power plant can also be considered. Nevertheless, the chosen annealing temperature should be such that it should affect neither the structural integrity nor the initial microstructure of the components.

Therefore, for the identification of microstructural processes responsible for the recovery of the mechanical properties, we performed quantitative TEM investigations on EUROFER97 irradiated

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to 15 dpa at 330 °C and subjected to PIA heat treatment at 550 °C for 3 h.

2. Experimental

The EUROFER97 steel plate (heat 83,697) investigated in the present study was produced by Böhler Austria GmbH with a chemical composition of 8.91 Cr, 1.08 W, 0.48 Mn, 0.20 V, 0.14 Ta, 0.006 Ti, 0.12 C (wt.%, Fe balance) [2]. The material was austenitized at 980 °C for 0.5 h and subsequently tempered at 760 °C for 1.5 h in air by the manufacturer prior to the delivery. Owing to its chemical composition and processing heat treatments, EUROFER97 manifests a complex ferritic-martensitic microstructure with a high density of interfaces (i.e. low- and high-angle grain boundaries), precipitates and dislocations [27].

The “Wissenschaftlich-Technische Zusammenarbeit 01/577” (WTZ) [25] irradiation experiments were performed on miniaturized Charpy impact samples of EUROFER97 in BOR-60 fast reactor of SSC RIAR in Dimitrovgrad under following condition: fast neutron dose of 15 dpa, neutron flux of $1.8 \times 10^{19} \text{ m}^{-2} \text{ s}^{-1}$ ($>0.1 \text{ MeV}$) and irradiation temperature of 330 °C. The acquired batches of both irradiated and post-irradiation annealed (at 550 °C with a temperature ramp of $\sim 2.9 \text{ K/min}$ and held for 3 h) impact samples were mechanically tested, and the detailed results are reported elsewhere [3,25,26].

Due to the limited amount of material and in order to avoid the influence of sample's magnetism on the TEM electron beam as well as to further minimize the radioactivity during microstructural investigations, a FEI Scios focused-ion beam (FIB) scanning electron microscope (SEM) was utilized in the present study [28]. Firstly, 1 mm disc samples were prepared out of undeformed parts of irradiated and post-irradiation annealed impact specimens by cutting thin slices and subsequent electrolytic polishing [5–7]. Thereafter, several TEM lamellae were in-situ lifted-out from the undeformed regions of the above-mentioned 1 mm discs and were attached onto the molybdenum TEM half-grid by Pt deposition. Finally, the lamellae were thinned down to electron transparency (to a thickness of 50–80 nm) with multiple passes of a low-energy ion beam. The FIB induced damage was minimized by optimizing cleaning pattern procedure parameters. In general, most FIB induced damage is similar to the irradiation-induced black dots. Therefore, to avoid any inappropriate estimation, the black dot features are excluded in this study. A high-resolution FEI Tecnai microscope operating at 200 kV equipped with a Gatan double-tilt holder and a high-angle annular dark-field (HAADF) scanning TEM (STEM) detector was employed for investigations. Thickness of the regions of interest was determined in TEM by employing electron energy loss spectroscopy (EELS) log-ratio method with about 10% uncertainty [29,30].

In general, $g \cdot b$ analysis (g is the diffraction vector) with weak beam dark-field (WBDF) technique is used to identify the dislocation loops Burgers vector b and calculate their density in irradiated material [31,32]. The underestimation of the loop density due to partial invisibility is resolved by using the statistical method for Burgers vector analysis [33]. The resolution limit for dislocation loops in terms of diameter was about 1 nm, and thus, it led to an underestimation of defect density with statistical error of about 20%. Additionally, bright-field (BF) and HAADF-STEM technique was used to quantify and characterize the microstructure. STEM offers advantage over conventional WBDF technique by significantly reducing the elastic contrast in the background by smearing out the thickness contrast. Since in the in-focus condition the voids or cavities are not visible, standard TEM-BF through-focal series technique was employed for their investigations [32]. A through-focal series under $+1.5 \mu\text{m}$ over-focus and $-1.5 \mu\text{m}$ under-

focus conditions provided optimum cavities contrast in the complex EUROFER97 microstructure.

3. Results

3.1. Dislocation loops and lines/networks evolution

Detailed microstructural characterization of EUROFER97 after WTZ irradiation (15 dpa, 330 °C) has been carried out several times [4–8]. An example of the representative microstructure under inverted WBDF and HAADF-STEM conditions are shown in Fig. 1a and Fig. 2a, respectively. As evident, under both conditions, the irradiated microstructure manifests homogenous distribution of the dark-contrast dislocation loops. Furthermore, in WBDF condition (Fig. 1a), the loops exhibit strong double-arc contrast and elliptical shape. These are a result of their projection, as loops habit planes are inclined to the camera viewing screen [34]. The appearance of double-arc contrast is a characteristic of interstitial type dislocation loops [35]. For loops quantitative analysis, several images were acquired and analysed under both WBDF and HAADF-STEM conditions. For example, the WBDF conditions $g(4.1 \text{ g})$ with $g = \{002\}$ diffraction vector and $g(6.1 \text{ g})$ with $g = \{110\}$ diffraction vector were analysed, resulting in the loop densities of $3.3 \times 10^{21} \text{ m}^{-3}$ and $4.0 \times 10^{21} \text{ m}^{-3}$, and mean sizes of 4.4 nm and 6.2 nm, respectively. Here, the major axis of an elliptical loop was taken as a measure of its size. By applying the loops invisibility criteria ($g \cdot b = 0$) and the statistical method for Burgers vector determination [33], the visible loops were found to be of both $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ types, with 27% and 73% fraction, respectively. Moreover, upon employing statistical method [33] in two different size regimes (i.e. above and below the loops mean size), the smaller loops are mostly found to be of $\frac{1}{2}\langle 111 \rangle$ type and the larger ones are dominantly $\langle 100 \rangle$ type. On the basis of this WBDF analysis, the total loop density and loop mean size were determined to be $6.4 \times 10^{21} \text{ m}^{-3}$ and 6.2 nm, respectively. Furthermore, from HAADF-STEM investigations, the total visible loop density in irradiated state was estimated to be about $12.1 \times 10^{21} \text{ m}^{-3}$, with the mean sizes of 7 nm. Clearly, using HAADF-STEM technique the estimated visible loops density is approximately two times higher than that measured via WBDF technique. This can be associated to the fact that all HAADF-STEM micrographs are taken close to the zone-axis orientations. And therefore, several excited Bragg's diffracted beams are captured by the HAADF-STEM detector in comparison to the few conditions used during WBDF analyses [36]. Furthermore, though HAADF-STEM technique provides atomic (Z) contrast, solute clusters or small precipitates that are generally visible under atom probe tomography (APT) are beyond STEM resolution [5,8]. As a result, the estimated total loop density using HAADF-STEM mode technique is higher.

In order to investigate the effect of PIA, similar investigations were performed on lamellae prepared from post-irradiation annealed samples. Fig. 1b and Fig. 2b present the representative microstructure after PIA under WBDF and HAADF-STEM conditions, respectively. Clearly, in contrast to the irradiated state, the annealed microstructure manifests a significantly lower dislocation loops density along with the frequent observations of dislocation lines/networks. Interestingly, some of these dislocation lines are found to be highly curved (Fig. 2b), giving impression that they might be a part of the large coarsened dislocation loops. Such large loops can be cut by the TEM lamellae surface and will be seen in the micrographs as the curved dislocation lines. Using line-intercept method, the observed dislocations density is found to be of the order of 10^{22} m^{-2} , which is qualitatively higher than that of the irradiated state. It is noteworthy that the quantitative estimation of the dislocations density in irradiated state was not possible due to the presence of the high density of dislocation loops. For

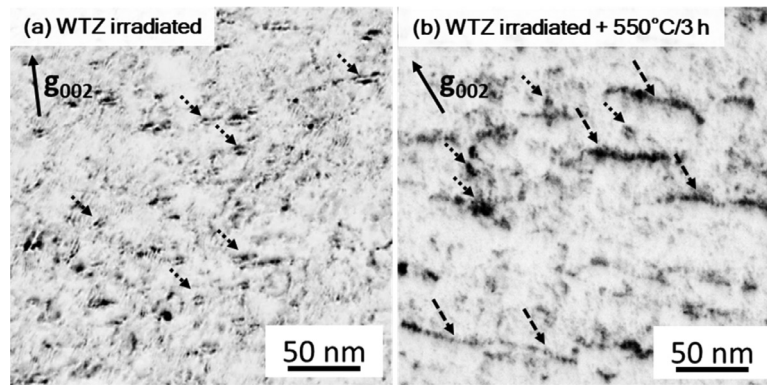


Fig. 1. Inverted contrast WBDF micrographs taken under $g(4.1\text{ g})$ diffraction condition with $g = \{002\}$ diffraction vector present representative microstructures of (a) WTZ irradiated (15 dpa, 330 °C) and (b) post-irradiation annealed (550 °C/3 h) EUROFER97 samples. The uniformly-distributed double-arc elliptical shaped dislocation loops in irradiated state (marked by dotted arrow) appear to have annealed out upon PIA. In addition, dislocation segments (marked by dashed arrow in (b)) are more commonly observed after PIA.

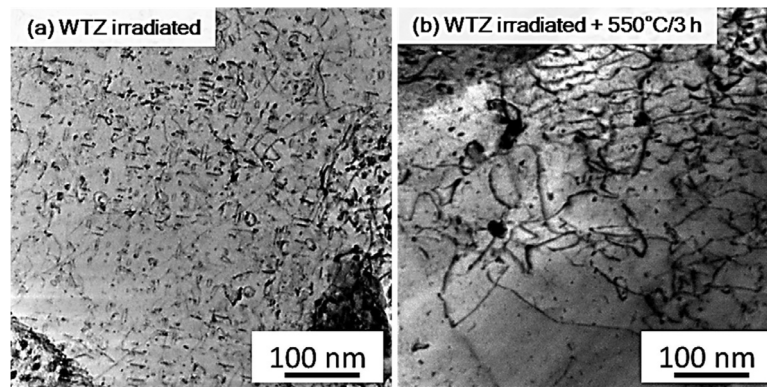


Fig. 2. Inverted-contrast HAADF-STEM micrographs taken close to the $\langle 110 \rangle$ zone-axis orientation present a representative microstructure of a (a) WTZ irradiated (15 dpa, 330 °C) and (b) post-irradiation annealed (550 °C/3 h) EUROFER97 samples. As evident, in contrast to the irradiated state, a significant reduction in dislocation loops density and increment in dislocation lines/networks density are apparent upon PIA.

Table 1

Summary of quantitative data of the dislocation loops in the WTZ irradiated and post-irradiation annealed (550 °C/3 h) EUROFER97 samples.

Condition	Technique	Total loop density ($\times 10^{21}\text{ m}^{-3}$)	Loop diameter (nm)
WTZ irradiated	WBDF	6.4 ± 1.9	6.2 ± 0.6
	HAADF-STEM	12.1 ± 0.7	7 ± 0.8
WTZ irradiated + 550 °C/3h	WBDF	2 ± 0.6	8 ± 0.6
	HAADF-STEM	4.1 ± 0.4	13 ± 0.9

loops quantitative analysis in annealed state, identical WBDF conditions, $g(4.1\text{ g})$ with $g = \{002\}$ diffraction vector and $g(6.1\text{ g})$ with $g = \{110\}$ diffraction vector were analysed, resulting in visible loop densities of $1.3 \times 10^{21}\text{ m}^{-3}$ and $1.1 \times 10^{21}\text{ m}^{-3}$, and mean sizes of 5.2 nm and 9.3 nm, respectively. By applying the loops invisibility criteria and the statistical method for Burgers vector determination [33], the visible loops were still found to be of both $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ types. Due to the difficulty in visualizing the large coarsened dislocation loops, the corresponding loops fraction was not estimated in the annealed state. On the basis of this WBDF analysis the total visible loop density and loop mean size after PIA were estimated to be $2.0 \times 10^{21}\text{ m}^{-3}$ and 8.0 nm, respectively. Furthermore, from HAADF-STEM investigations, the total visible loop density after PIA was estimated to be about $4.1 \times 10^{21}\text{ m}^{-3}$, with the mean sizes of about 13 nm. A summary of the quantitative data of the dislocation loops is listed in Table 1. Finally, upon comparison with the irradiated state, PIA results in a reduction of the total loops density by 69% and 66% from WBDF and HAADF-STEM analysis, respectively.

Additionally, the comparison of the dislocation loops size distribution obtained via WBDF and HAADF-STEM investigations after WTZ irradiation and PIA conditions are shown in Fig. 3. As evident, a clear reduction of the dislocation loops density as well as an obvious shift of their size distribution towards higher size is apparent upon PIA. This suggests that the loops mean size increase is not exclusively a result of smaller loops shrinkage/annihilation but is also associated to the actual loops coarsening upon PIA. However, though additional investigations are required for confirmations, the mechanisms such as loops shrinkage, annihilation, coarsening and their coalescence could be active and linked to the present observations.

3.2. Voids or cavities evolution

Upon neutron irradiation, vacancies are created in the damage cascades which can aggregate to form voids (empty clusters) or cavities (clusters of vacancies that contain helium). In the investigated EUROFER97, only very small amount of boron (about 10 wt. ppm) is present in the unirradiated state [37]. Thus, only little

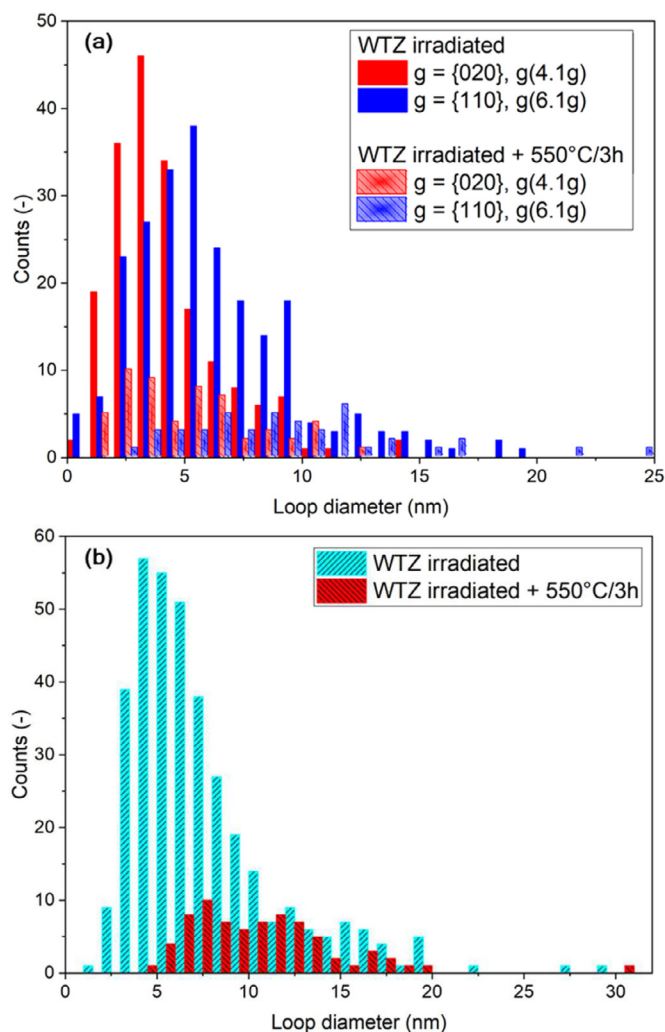


Fig. 3. Dislocation loops size distributions for the WTZ irradiated (15 dpa, 330 °C) and post-irradiation annealed (550 °C/3 h) EUROFER97 samples estimated via (a) WBDF and (b) HAADF-STEM techniques. A clear reduction in dislocation loops density and distribution shift towards higher size are apparent upon PIA. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

amount of He would have transmuted from Boron during WTZ irradiation (15 dpa, 330 °C) [6]; consequently, some cavities along with mostly voids are expected to exist in the investigated EUROFER97. Note that for better description hereafter we have used the word “cavities” to represent both voids and cavities. Investigation of cavities formation after WTZ irradiation has also been reported before [4]. However, the statistics of the previous measurement were poor; and hence, additional investigations and analyses were carried out to reconfirm previous results. By using standard technique of TEM-BF through-focal series, several micrographs were acquired. In general, the irradiated microstructure exhibits an inhomogeneous distribution of cavities. Fig. 4a and b show cavities distribution in a region with relatively uniform density. As typical, the contrast of the cavities changed from a white dot with a black fringe in an under-focused image (Fig. 4a) to a dark dot with a white Fresnel fringe in the over-focused image (Fig. 4b). Contrary to the regions with relatively uniform density, several cavity free areas that are hundreds of nanometers across were also observed. This inhomogeneity appears to be an effect of the material's inherent local chemical composition variation; for example, the boron distribution [38,39]. Using several micrographs, cavities size distribution

was determined and is shown in Fig. 5a. The mean cavity diameter (fringe edge-to-edge distance) is estimated to be 2.12 nm and the density is determined to be $3.02 \times 10^{21} \text{ m}^{-3}$. This measured cavity density is one order of magnitude higher than that reported before ($3.6 \times 10^{20} \text{ m}^{-3}$) [4] and comparable to those observed after ARBOR1 ($2.3 \times 10^{21} \text{ m}^{-3}$, mean diameter 1.6 nm) irradiation (32 dpa, 330 °C) [6].

In order to investigate the effect of PIA, similar investigations were performed on the post-irradiation annealed samples (WTZ irradiated + 550 °C/3 h). Fig. 4c and d show cavities distribution in a region comparable to that shown for the irradiated case with relatively uniform density. Again, using several micrographs, cavities distribution was determined for the post-irradiation annealed samples, which is shown in Fig. 5b. Here, their mean diameter is estimated to be 2.9 nm and their density is determined to be $1.5 \times 10^{21} \text{ m}^{-3}$ after PIA. Therefore, upon comparison with the irradiated state, an appreciable reduction in the cavities density (~50%) and increment in their size (~36%) is apparent upon PIA.

4. Discussion

In this work, TEM investigations on the neutron-irradiated and subsequently annealed EUROFER97 samples were carried out to understand the effect of annealing on irradiation induced defects. Both WBDF and HAADF-STEM investigations reveal a reduction of dislocation loops density with a simultaneous increase in their mean size and formation or extension of dislocation lines/networks upon PIA. Additionally, a reduction of cavities density with increment in their mean size was observed. Though not as much for RAFM steels, similar PIA observations have also been reported before for other irradiated austenitic steels [16–24], which is associated with the recovering of the material's hardening and embrittlement behaviour. These observations have usually been linked to the mechanisms that are either vacancy/interstitial diffusion controlled or depend on the elastic forces between defects themselves [17].

The reduction of the dislocation loops density upon PIA can be a result of their 1) annihilation and/or 2) coarsening. The dominance of one phenomenon over the other defines the overall loops size evolution. Loops shrinkage and their disappearance have been reported before upon annealing irradiated pure α -iron, Fe-9%Cr alloy and austenitic steels [12,21,24,40,41]. It is known that excess vacancies exist in matrix at elevated temperatures due to the increase in the thermal equilibrium vacancies. And therefore, at the chosen annealing temperature, vacancies can move in the matrix and flow into sinks, such as surface, grain boundaries, dislocation tangles and dislocation loops [21]. Concerning loops, vacancies are attracted to its peripheral high compressive strain field region [42], since the vacancy concentration at the loop is lower than the thermal equilibrium concentration [43]. The incoming vacancies will then annihilate with the loop's peripheral self-interstitial atoms, which firstly leads to their shrinkage (progressive reduction of the loops size) and finally, to their complete annihilation. Indeed, authors recent in-situ annealing investigations on irradiated EUROFER97 [44] directly revealed vacancy diffusion mediated shrinkage phenomenon for both $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ type isolated loops. However, shrinkage rate was found to be both loops size and type dependant. Fast shrinkage was mostly observed for the small $\frac{1}{2}\langle 111 \rangle$ type loops which were present in the close proximity of the dislocation tangles, indicating tangles to be the source and sink for point defects (i.e. vacancies and interstitials) [44]. And therefore, it appears that loops in the population can behave as an isolated loop and shrink/annihilate independently, if sufficient external sources and sinks (e.g. surfaces, grain boundaries, cavities and dislocation tangles) are present in their proximity [17,43]. Besides, the large elastic misfit associated with an interstitial defect

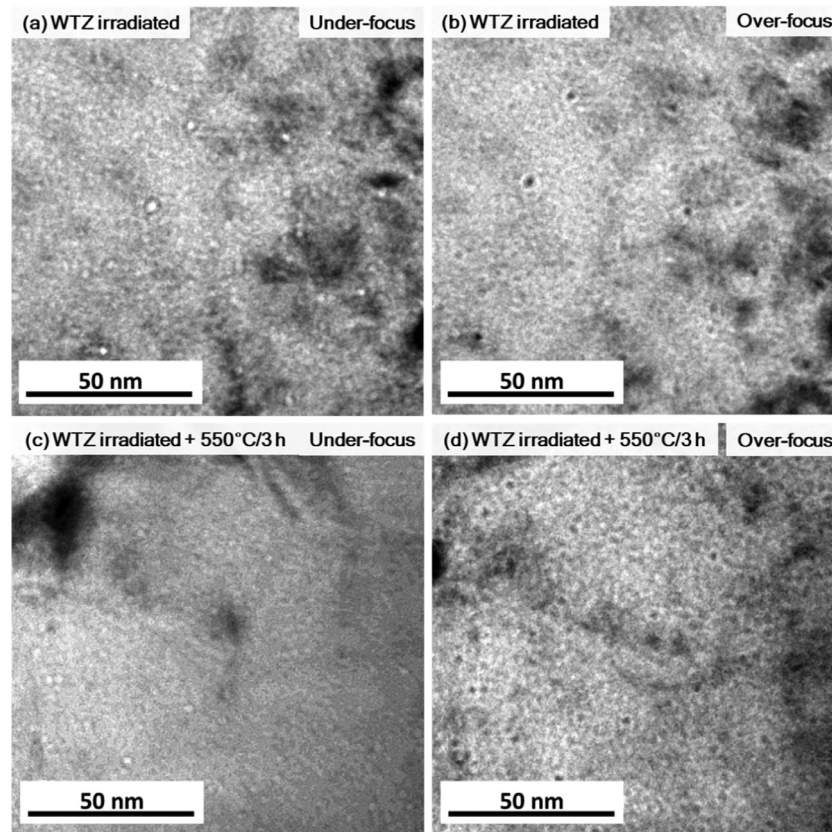


Fig. 4. Typical BF-micrographs present cavity distribution in a (a, b) WTZ irradiated (15 dpa, 330 °C) and (c, d) subsequently annealed (550 °C/3 h) EUROFER97 samples. (a/c taken at under-focus $-1.5 \mu\text{m}$ condition and b/d taken at over-focus $+1.5 \mu\text{m}$ condition).

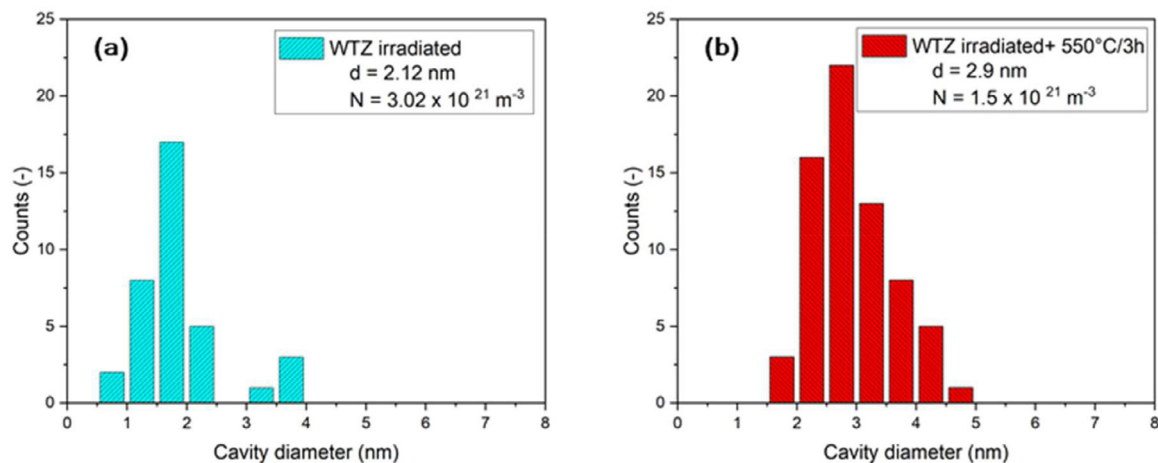


Fig. 5. Cavities size distribution along with their average size d and volume density N for the (a) WTZ irradiated (15 dpa, 330 °C) and (b) post-irradiation annealed (550 °C/3 h) EUROFER97 samples. An appreciable reduction in the cavities density and increment in their size is apparent upon PIA. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

can cause dislocations to attract interstitials more strongly than vacancies [45]. Consequently, interstitial loops/clusters can also be absorbed by the present dislocations. Similar observations of the Frank loops shrinkage along with their annihilation upon interacting with complex dislocation networks was reported by Holmes et al. [16] upon annealing neutron-irradiated austenitic stainless steel. However, though loops annihilation can explain the reduction of their density, some additional mechanisms could also operate simultaneously that could be linked to the coarsening phenomenon.

In general, two diffusion-controlled mechanisms of loops coarsening: 1) growth of individual loops via absorption of highly mobile interstitials and 2) merging of two or more dislocation loops can occur upon annealing. Moll et al. [46], upon annealing irradiated pure α -iron with low initial dislocation density at 577 °C, reported significant coarsening of loops with the emission of vacancies (absorption of highly mobile interstitials). This coarsening is accompanied by the reduction of dislocation loops density, due to their coalescence [46]. In general, the coalescence or merging of loops befalls once they are sufficiently large to interact and no

other external sources and sinks exist nearby but loops themselves [17,43,46]. The loops interaction is also known to be both loops size and type dependant. For example, the emission rate of vacancies and absorption rate of interstitials, is greater for the large loops than for the relatively small loops [46]. So, in order to conserve the total number of vacancies/interstitials, the larger loops grow at an expense of smaller ones [17,19]. This coarsening is similar to the particle coarsening (Oswald ripening) in the material where the total amount of solute remains constant. Additionally, since $\langle 100 \rangle$ type dislocation loops is generally larger it has been shown to coarsen faster than the $\frac{1}{2}\langle 111 \rangle$ counterparts [47]. Furthermore, for BCC iron, Masters [48] proposed and Xu et al. [49] via kinetic Monte Carlo simulation showed that two small $\frac{1}{2}\langle 111 \rangle$ type dislocation loops can coalesce to form one larger $\langle 100 \rangle$ type dislocation loop or one larger $\frac{1}{2}\langle 111 \rangle$ type dislocation loop. Arakawa et al. [15] via in-situ annealing investigations showed similar results for irradiated pure α -iron. Besides, interstitial loops in pure α -iron are known to be mobile [10,14,50,51]. Therefore, for the glissile loops, a further mechanism of coarsening can arise as a result of the direct elastic interaction force between adjacent loops [10,52]. The forces exerted by one loop on the other can make them move together by glide and/or climb, which can lead to their coalescence [10].

In contrast to pure α -iron, different alloying elements (Cr, Ni, Si, Mn and P) are known to decorate loops periphery in irradiated EUROFER97 in order to relax associated dilatational-strain field [8,12]. This segregation is known to suppress dislocation loops evolution upon annealing. For example, Arakawa et al. [12] showed suppressed loops mobility as well as their shrinkage by Cr addition in irradiated pure α -iron during in-situ annealing. Moreover, preferential segregation based on the loops size and type could have significant suppressing effect on the loops (larger $\langle 100 \rangle$ type) shrinkage behaviour [44]. Yoshida et al. [53] additionally suggested that Cr at dislocation loops could reduce the absorbing reaction between dislocation loops and interstitials, which could suppress their coarsening behaviour. Lastly, elastic forces governed glide plus climb mechanism is more likely to be active in high purity metals (e.g. pure α -iron) than in their alloyed counterparts such as EUROFER97, where loops are generally sessile. In spite of all this, substantial recovery of the irradiated EUROFER97 microstructure is observed upon PIA. Therefore, from present investigations it can be concluded that the loops annealing behaviour of the irradiated EUROFER97 is primarily vacancy/interstitial diffusion driven and it falls somewhere between the two extremes of isolated loops shrinkage/annihilation and their coarsening, depending upon the absence or presence of the external sources and sinks. Similar loops annealing behaviour has also been reported before for irradiated austenitic steel [21].

Concerning dislocation lines/networks, both ferritic-martensitic (T91) [54] and cold-worked austenitic stainless steels [55] have been shown to have undergone reduction in dislocation lines/networks density upon irradiation, prominently at higher doses. Likewise, in comparison to the unirradiated state, irradiated EUROFER97 also manifest lower dislocation lines/networks density. This has been generally attributed to the irradiation induced dislocations-climb (by absorbing point defects) which leads to the mutual annihilation of the line dislocations [55]. However, prominent increment in dislocation lines/networks density is apparent upon PIA EUROFER97. It is known that the loops interaction during coarsening/coalescence process could lead to the formation of dislocation lines/networks [20,46]. Fig. 6 shows example of the possible loop-loop and dislocation-loop interactions, along with the newly formed dislocation lines/networks in a post-irradiation annealed EUROFER97 sample. Here, some of the dislocations are found to be highly curved which gives impression that they might be a part of large coarsened dislocation loops.

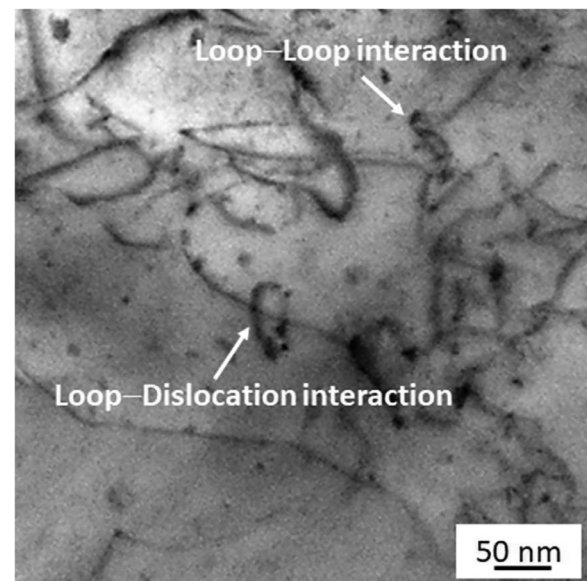


Fig. 6. HAADF-STEM micrograph reveals possible loop-loop and loop-dislocation interactions, along with the newly formed dislocation lines/networks, in a post-irradiation annealed EUROFER97 sample.

Indeed, Klimenkov et al. [11] found “curved dislocation lines” in the neutron-irradiated EUROFER97 (16.3 dpa, 400–450 °C) that were a part of the large dislocation loops. It should be mentioned that recognizing such coarsened loops amongst actual dislocation lines/networks is quite complicated. Thus, the loops actual mean size as well as their density could be higher than the given measured values for annealed state. Besides, interstitial loops/clusters that are absorbed by the background dislocations could further lead to their extension. Masters [48] for ion-irradiated iron proposed that $\langle 100 \rangle$ loops upon interaction with glide dislocations can convert into glissile dislocations. And as appreciable dislocation activity was observed in our in-situ annealing investigations [44], such reaction could be responsible for the reduction of loops density. Similar observations of dislocation lines/network formation were also reported before upon annealing irradiated pure α -iron [46] and austenitic steels [18,20,23].

In the present study, analogous to the loops behaviour, upon PIA cavities density reduction and their mean size increment was also observed. Similar observations have also been reported before upon annealing He-ion irradiated RAFM steel at 550 °C [56]. However, though in the investigated EUROFER97 the cavities are mostly vacancy clusters, along with few that can be He stabilized, similar mechanisms as those generally proposed for the He bubbles coarsening might be operating in the present case. The two qualitative coarsening mechanisms that explain such observations upon annealing are: 1. Vacancy or their cluster migration and coalescence and 2. Ostwald ripening. The activation energy of Ostwald ripening mechanism is expected to be much higher than that of the migration and coalescence one [57]. Therefore, vacancy migration and coalescence and Ostwald ripening mechanisms are expected to be dominant at relatively low and high temperatures, respectively [57]. And since, local vacancies around cavities are easier source, migration and coalescence mechanism appears to be the main possible mechanism of the cavities growth. Additionally, cavities coarsening is also substantially complicated in the presence of and close to precipitates, dislocations and grain boundaries [57]. And as significant dislocation activity was observed during recent in-situ annealing experiments [44], the cavities sweeping by dislocation motion can also contribute to their coarsening.

Since irradiation induced defects (for e.g. dislocation loops, cavities, segregation, solute clusters and precipitates) are known to act as a barrier for dislocations motion [8], healing of these defects during PIA could lead to the recovery of the mechanical properties [3,58]. Indeed, previous PIA investigations on EUROFER97 has led to a nearly complete recovery of the low temperature hardening [3,25,26]. Present TEM investigations also support these results as substantial recovery of the irradiated microstructure, in terms of the reduction of the loops and cavities density is observed upon PIA. Nevertheless, it still remains unclear why the hardening recovered almost fully even though the microstructure is not fully recovered. The possible reason lies in understanding that how other significant irradiation-induced hardening contributors (such as loops, segregation and solute clusters [59,60] that are beyond TEM resolution) evolve during PIA and how they alter total hardening superimpositions. Therefore, additional investigations using high-resolution techniques like atom-probe tomography (APT) are required to complete characterization of the post-irradiation annealed EUROFER97 microstructure.

5. Conclusion

In this work, TEM investigations on the neutron-irradiated and subsequently annealed EUROFER97 samples were carried out to understand the effect of annealing on irradiation induced defects. Both WBDF and HAADF-STEM investigations reveal a reduction of dislocation loops density with a simultaneous increase in their mean size and formation of dislocation lines/networks upon PIA. Additionally, a reduction of cavities density with increment in their mean size was observed. The loops annealing behaviour is primarily vacancy/interstitial diffusion driven and it falls somewhere between the two extremes of isolated loops shrinkage/annihilation and their coarsening. Out of the two phenomena, the one that transpires depend on the presence or absence of the external sources and sinks in the loop's neighbourhood. These competing mechanisms define the overall loops size distribution evolution upon PIA. Loops interaction during coarsening/coalescence process also leads to the formation of dislocation lines/networks. Additionally, vacancy or their cluster migration and coalescence could be the main mechanism for cavities coarsening. However, to understand different operating mechanisms in detail systematic isochronal and isothermal in-situ annealing experiments are required. Nevertheless, the observed microstructural recovery upon PIA explains the recovery of mechanical properties by healing irradiation induced defects.

6. Credit author statement

Ankur Chauhan: Supervision, Investigation, Formal analysis, Writing - Original Draft; Qian Yuan: Investigation, Writing - Review & Editing; Christian Dethloff: Writing - Review & Editing; Ermile Gaganidze: Supervision, Funding acquisition, Writing - Review & Editing; Jarir Aktaa: Writing - Review & Editing

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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